

Smartphone-enabled functions

SenseGiz STAR is supported by a free app for Android and iOS devices. The user's fitness data in terms of number of steps taken, amount of calories burnt and total distance covered, and sleep quality in terms of light sleep, deep sleep, total sleep and interruptions can be viewed on the app. Other features like user-defined gesture control, setting alert notifications for crashes/falls and defining emergency contacts can also be performed using this app. STAR regularly, and automatically, synchronises with the phone app via BLE and even displays call, text and app notifications on its OLED screen.

Hurdles in syncing with app and cloud

To ensure the feature that displays time on the screen like a digital watch or clock works up to the mark, developers needed to ensure that this time display is synchronised with Greenwich Mean Time (GMT) and the local time zone through phone synchronisation.

On the cloud side, the user needs to ensure a regular synchronisation mechanism of the gadget with the mobile app. But at the same time, all calculations on the app side need to be performed within one or two seconds, so that information and guidance can be communicated to the user in real-time.

The issue was approached by using a lean-sync mechanism with the cloud at regular intervals. The team minimised the usage of the central

Specifications

Device	: STAR
Brand	: SenseGiz
Founders	: Abhishek Latthe, CEO, and Apurva Shetty, COO
Models	: Two designs: clip-on and wristband Four colours: classy black, flawless white, bubbly blue and perky pink
Pricing	: ₹ 6000
Website	: www.sensegiz.com/star
Contact	: info@sensegiz.com

processing unit (CPU) of the smartphone by the app for all calculations, thereby avoiding battery wastage.

Challenging algorithm

The major challenge faced by the team while designing the product was to eliminate any kind of false crash alarms that would negatively affect the product functionality and reliability.

Another challenge was to differentiate between deep sleep and light sleep as the device is supposed to work as a sleep-quality tracker.

The next goal was to precisely count the number of steps taken by the user during a walk. Also, the device is supposed to accurately interpret hand gestures to perform different operations associated with it by the user.

The fall/crash-detection algorithm takes into account what is happening before, during and after a fall/crash event has taken place. For example, forces associated with high-impact and rapid body movements during a fall and movements of the user after the event are monitored to make a decision on the event.

After several iterations, the team was finally able to come up with well-built algorithms that can distinguish between a fall and a crash, would not report any false crash alarms, monitor the type and duration of sleep, count the number of steps covered and interpret hand gestures in an effective manner. SenseGiz has

also applied patent for the fall/crash algorithm, which they have developed.

How is STAR different

There are several fitness bands of varying functionalities and prices available in the market these days. Why choose STAR over other products? The team explains, "Safety is the main feature we focus on and this is what differentiates our product from our competitors."

The device automatically detects falls and crashes, even the severity of these, and an alarm in the preset phone number(s) automatically goes off in case of an emergency. This alarm is more likely to get noticed by another person rather than a simple text message (SMS, in most cases) featured by some gadgets.

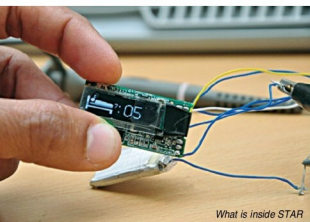
STAR also has a panic button that can be pressed when physical danger is anticipated or the user needs immediate help. The logic behind this is that, pressing this panic button is easier than searching for the phone and calling someone, especially when time is a critical factor.

Now, the design challenge arises; what if the panic button is pressed by mistake? In order to activate a panic alert, this button should be quickly pressed twice, so chances of false alerts are less. Also, there is a user-defined countdown timer on the app that can disable sending of alerts if there is no emergency.

Multiple features like safety, fitness and UI screen displaying notifications come together at a competitive price and no other wearable device offers all these features in a single product.

From idea to market

The development of STAR from idea into a fully-functional product took approximately one year, says the SenseGiz team. The clip version of the device is already being sold directly to customers or through retailers in more than ten countries across the globe. ●



What is inside STAR